

Cognitive Performance OS

AI Trainer for Structured Thinking Under Pressure

Hackathon – Investor Grade Concept Document

1. The Real Problem

- Students know concepts but cannot structure thoughts under pressure.
- They ramble when interrupted in viva or interviews.
- They fail not due to knowledge gap but cognitive structure gap.
- Existing tools train grammar, not thinking architecture.

2. Our Solution

- Cognitive Structure Heatmap
- Mid-Interruption Recall Engine
- Idea Compression System
- Adaptive Pressure Mode
- Slide-to-Speech Alignment Analyzer

3. System Flow Diagram

User Selects Mode
AI Asks Structured Question
User Responds (Voice/Text)
Mid-Answer Interruption Triggered
Compression Challenge Activated
Cognitive Heatmap Generated
Dashboard Score Updated

4. System Architecture

Frontend	React Native + NativeWind
Backend	Node.js (Recommended for real-time handling)
Database (Primary)	PostgreSQL (Structured scoring model)
Database (Optional Logs)	MongoDB (Flexible interaction logs)
AI Layer	OpenAI GPT-4 API
Speech-to-Text (Optional)	OpenAI Whisper API

5. Tech Stack Justification

- React Native ensures cross-platform scalability.
- NativeWind enables fast UI styling.
- Node.js handles real-time interruptions efficiently.
- PostgreSQL ensures structured performance scoring.
- MongoDB optional for unstructured speech logs.
- GPT-4 API enables advanced reasoning and structure evaluation.

6. Initial Target Segment (V1 Focus)

- Final-year engineering students.
- Campus interview preparation.
- Viva and project defense training.

7. Positioning Statement

Cognitive Performance OS is not a grammar tool or mock interview bot. It is an AI-powered cognitive structuring trainer that builds logical thinking and interruption resilience under real-world pressure.